MA1522 Linear Algebra for Computing

Lesson 6: Properties of Determinants

Announcement: Online Quiz 1

- **Quiz 1 covers Chapter 1**, Sections 1.1–1.5: linear systems, row-echelon forms, elementary row operations, and Gaussian and Gauss–Jordan elimination.

- It opens **Saturday 29 August, 0:00** and is due **Sunday 30 August, 23:59**.

Time limit is 30 mins.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson6` (e.g. `A1234567Z_lesson6.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 6**.
- Deadline: today, **10:30 AM**.
- If you do not know how to upload your submission, come to ask me after class.

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Basic rules:

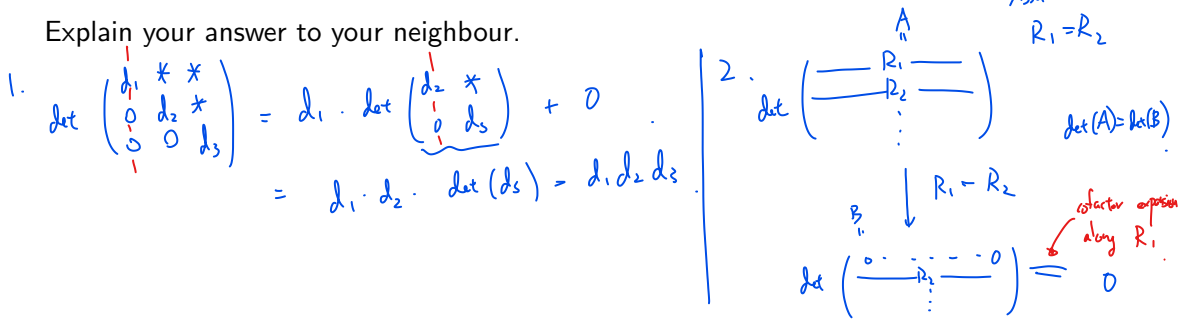
- Please (mainly) ask the questions related to the course.

Example 1

True or false?

1. For an upper triangular matrix, the determinant is equal to the product of the diagonal entries.
2. If two rows of a square matrix are identical, then the determinant is zero.

Explain your answer to your neighbour.



Section 2.10 Determinant and Invertibility

Theorem

A square matrix A is invertible if and only if $\det(A) \neq 0$.

So the list of **equivalent statements for invertibility** from Section 2.6 gains one more entry. For a square matrix A of order n, the following say the same thing:

(i) A is invertible;

(v) the reduced row-echelon form of A is I_n;

(vii) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;

(viii) for any $\mathbf{b}$, the system $A\mathbf{x} = \mathbf{b}$ has a unique solution;

(ix) $\det(A) \neq 0$.

A square matrix that is not invertible is called **singular**.

Question 1

Let $x \in \mathbb{R}$ and

$$A = \begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix}.$$

1. Compute $\det(A)$. Compute by definition. I recommend expansion by cofactor.
2. Find all x for which A is singular.
   not invertible.
3. For each such x, find a nonzero solution of $A\mathbf{x} = \mathbf{0}$.

Question 1 — solution: parts (1) and (2)

(1) $\det \begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix} = (-1)^{1+1}(-x) \cdot \det \begin{pmatrix} -x & 1 \\ -5 & 4-x \end{pmatrix} + (-1)^{1+2} \cdot 1 \cdot \det \begin{pmatrix} 0 & 1 \\ 2 & 4-x \end{pmatrix} + \underbrace{(-1)^{1+3} \cdot 0 \cdot \begin{pmatrix} -x \end{pmatrix}}_{0}$

$= (-x) \cdot \left((-x)(4-x) - 1 \cdot (-5) \right) - \left(0 \cdot (4-x) - 1 \cdot 2 \right)$

$= -x \cdot \left(x^2 - 4x + 5 \right) + 2$

$= -x^3 + 4x^2 - 5x + 2$.

(2) Find all x for which A is singular $\iff \det(A) = 0$.

With calculator, "find roots", $\det(A) = 0 \iff x = 1, 2$.

$\det(A) = -(x-1)^2(x-2)$.

Question 1 — solution: part (3)

For $x=1, 2$. find a non-trivial sol to $\begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix} \cdot \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

$x=1$

$$\left(\begin{array}{ccc|c} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 2 & -5 & 3 & 0 \end{array}\right).$$

$\downarrow$ ref

$$\left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

general sol.

$$\begin{pmatrix} t \\ t \\ t \end{pmatrix} \quad t \in \mathbb{R}.$$

$$v = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Remark (diagonalization / eigen vector / eigen value)

$A = B - xI_3$ $\quad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 2 & -5 & 4 \end{pmatrix}$.

$x=1$.

$$A \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Updownarrow$

$$(B - 1 I_3) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

$\Updownarrow$

$$B \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

$x=2$

$$A w = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$\Updownarrow$

$$B \cdot w = 2w.$$

Section 2.10 Determinant of a Product, a Scalar Multiple, an Inverse

Product

For square A, B of the same size,

$$\det(AB) = \det(A)\det(B),$$

and by induction
$\det(A_1 \cdots A_k) = \det(A_1) \cdots \det(A_k)$.

Scalar multiple

For A of order n and $c \in \mathbb{R}$,

$$\det(cA) = c^n \det(A).$$

There is no rule for $\det(A + B)$.

Inverse

If A is invertible, then

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Transpose (Section 2.8)

$$\det(A^T) = \det(A).$$

Example 2

Let A, B and P be square matrices of the same size, with P invertible. True or false?

1. $\det(AB) = \det(BA)$.
2. $\det\big(PA(P^{-1})^T\big) = \det(A)$.

1. $\det(AB) = \det(A)\det(B) = \det(B)\det(A) = \det(BA)$

2. $\det\big(PA(P^{-1})^T\big) = \det(P)\det(A)\det\big((P^{-1})^T\big) = \det(A)$.

$\det\big((P^{-1})^T\big) = \det(P^{-1}) = \det(P)^{-1}$

Section 2.10 The Adjoint

Let A be a square matrix of order n, with cofactors $A_{ij} = (-1)^{i+j}\det(M_{ij})$. The **adjoint** of A, written $\text{adj}(A)$, is the $n \times n$ matrix whose (i,j)-entry is the (j,i)-cofactor of A:

$$\text{adj}(A) = \begin{pmatrix} A_{11} & A_{21} & \cdots & A_{n1} \\ A_{12} & A_{22} & \cdots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \cdots & A_{nn} \end{pmatrix}$$

— the cofactor matrix, transposed.

Adjoint formula

$$A\,\text{adj}(A) = \det(A)\,I_n = \text{adj}(A)\,A.$$

Inverse, when $\det(A) \neq 0$

$$A^{-1} = \frac{1}{\det(A)}\,\text{adj}(A).$$

Question 2

Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}.$$

1. Compute the nine cofactors of A, and hence $\mathrm{adj}(A)$. You do not need to expand the nine 2×2 determinants.

2. Compute the $(1,1)$ entry of $A\,\mathrm{adj}(A)$, and explain why it is equal to $\det(A)$.

3. Compute the $(2,1)$ entry of $A\,\mathrm{adj}(A)$. Identify it with the determinant of a certain matrix, and explain why it is zero.

Question 2 — solution: part (1)

$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$ $\quad \left|\begin{matrix} a & b \\ c & d \end{matrix}\right| = \det\begin{pmatrix} a & b \\ c & d \end{pmatrix}$

(1)

$$\operatorname{adj}(A) = \begin{pmatrix} \left|\begin{matrix} 1 & 4 \\ 6 & 0 \end{matrix}\right| & -\left|\begin{matrix} 0 & 4 \\ 5 & 0 \end{matrix}\right| & \left|\begin{matrix} 0 & 1 \\ 5 & 6 \end{matrix}\right| \\ -\left|\begin{matrix} 2 & 3 \\ 6 & 0 \end{matrix}\right| & \left|\begin{matrix} 1 & 3 \\ 5 & 0 \end{matrix}\right| & -\left|\begin{matrix} 1 & 2 \\ 5 & 6 \end{matrix}\right| \\ \left|\begin{matrix} 2 & 3 \\ 1 & 4 \end{matrix}\right| & -\left|\begin{matrix} 1 & 3 \\ 0 & 4 \end{matrix}\right| & \left|\begin{matrix} 1 & 2 \\ 0 & 1 \end{matrix}\right| \end{pmatrix}^T = \begin{pmatrix} \left|\begin{matrix} 1 & 4 \\ 6 & 0 \end{matrix}\right| & -\left|\begin{matrix} 2 & 3 \\ 6 & 0 \end{matrix}\right| & \left|\begin{matrix} 2 & 3 \\ 1 & 4 \end{matrix}\right| \\ -\left|\begin{matrix} 0 & 4 \\ 5 & 0 \end{matrix}\right| & \left|\begin{matrix} 1 & 3 \\ 5 & 0 \end{matrix}\right| & -\left|\begin{matrix} 1 & 3 \\ 0 & 4 \end{matrix}\right| \\ \left|\begin{matrix} 0 & 1 \\ 5 & 6 \end{matrix}\right| & -\left|\begin{matrix} 1 & 2 \\ 5 & 6 \end{matrix}\right| & \left|\begin{matrix} 1 & 2 \\ 0 & 1 \end{matrix}\right| \end{pmatrix}$$

(2) Compute the $(1,1)$ entry of $A \operatorname{adj}(A)$.

$$\left(A \operatorname{adj}(A)\right)_{11} = 1 \cdot \left|\begin{matrix} 1 & 4 \\ 6 & 0 \end{matrix}\right| + 2 \cdot \left(-\left|\begin{matrix} 0 & 4 \\ 5 & 0 \end{matrix}\right|\right) + 3 \cdot \left|\begin{matrix} 0 & 1 \\ 5 & 6 \end{matrix}\right|.$$

This is the cofactor expansion for $\det(A)$ along the first row!

$$= \det(A).$$

Similar computation works for $(2,2)$, $(3,3)$ entries.

Question 2 — solution: part (2)

(3) $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$.

$$\operatorname{adj}(A) = \begin{pmatrix} \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} & -\begin{vmatrix} 2 & 3 \\ 6 & 0 \end{vmatrix} & \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} \\ -\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 3 \\ 5 & 0 \end{vmatrix} & -\begin{vmatrix} 1 & 3 \\ 0 & 4 \end{vmatrix} \\ \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 5 & 6 \end{vmatrix} & \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \end{pmatrix}$$

Compute $(2,1)$-entry of $A \cdot \operatorname{adj}(A)$. Interpret it as a determinant of a certain matrix. Explain why it is zero.

$$(A \operatorname{adj}(A))_{21} = 0 \cdot \begin{vmatrix} 1 & 4 \\ 6 & 0 \end{vmatrix} + 1 \cdot \left(-\begin{vmatrix} 0 & 4 \\ 5 & 0 \end{vmatrix}\right) + 4 \cdot \begin{vmatrix} 0 & 1 \\ 5 & 6 \end{vmatrix} = \det(M) = 0$$

If this is cofactor expansion along the first row of M.

$$M = \begin{pmatrix} 0 & 1 & 4 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}.$$

M has two identical rows.

Question 2 — solution: part (3)

Question 3

$\iff \det(A) \neq 0.$

Let A be an invertible square matrix of order $n \geq 2$. Show that

$$\det(\text{adj}(A)) = \det(A)^{n-1}.$$

Explain your proof to your classmates.

pf We use. $\det\left(A \cdot \text{adj}(A)\right) = \det(A) \cdot I_n = \det\begin{pmatrix} \det(A) & & 0 \\ & \det(A) & \\ 0 & & \ddots \\ & & & \det(A) \end{pmatrix} = \det(A)^n$.

$\det(A) \cdot \det\left(\text{adj}(A)\right)$

$\det(A) \cdot \det(A) \cdots \det(A)$ (n copies)

Question 3 — solution

In conclusion

$$\det(A) \cdot \det(\operatorname{adj}(A)) = \det(A)^n$$

Since $\det(A) \neq 0$, we obtain

$$\det(\operatorname{adj}(A)) = \det(A)^{n-1}$$

□

Challenge

Question 3 assumed A invertible, and the proof needed that: it divided by $\det(A)$.

Does $\det(\text{adj}(A)) = \det(A)^{n-1}$ still hold when A is singular?

Next lesson

- **Sections 3.1–3.3 Euclidean Spaces, Dot Product, and Linear Spans**

Please watch the video clips for these sections beforehand.